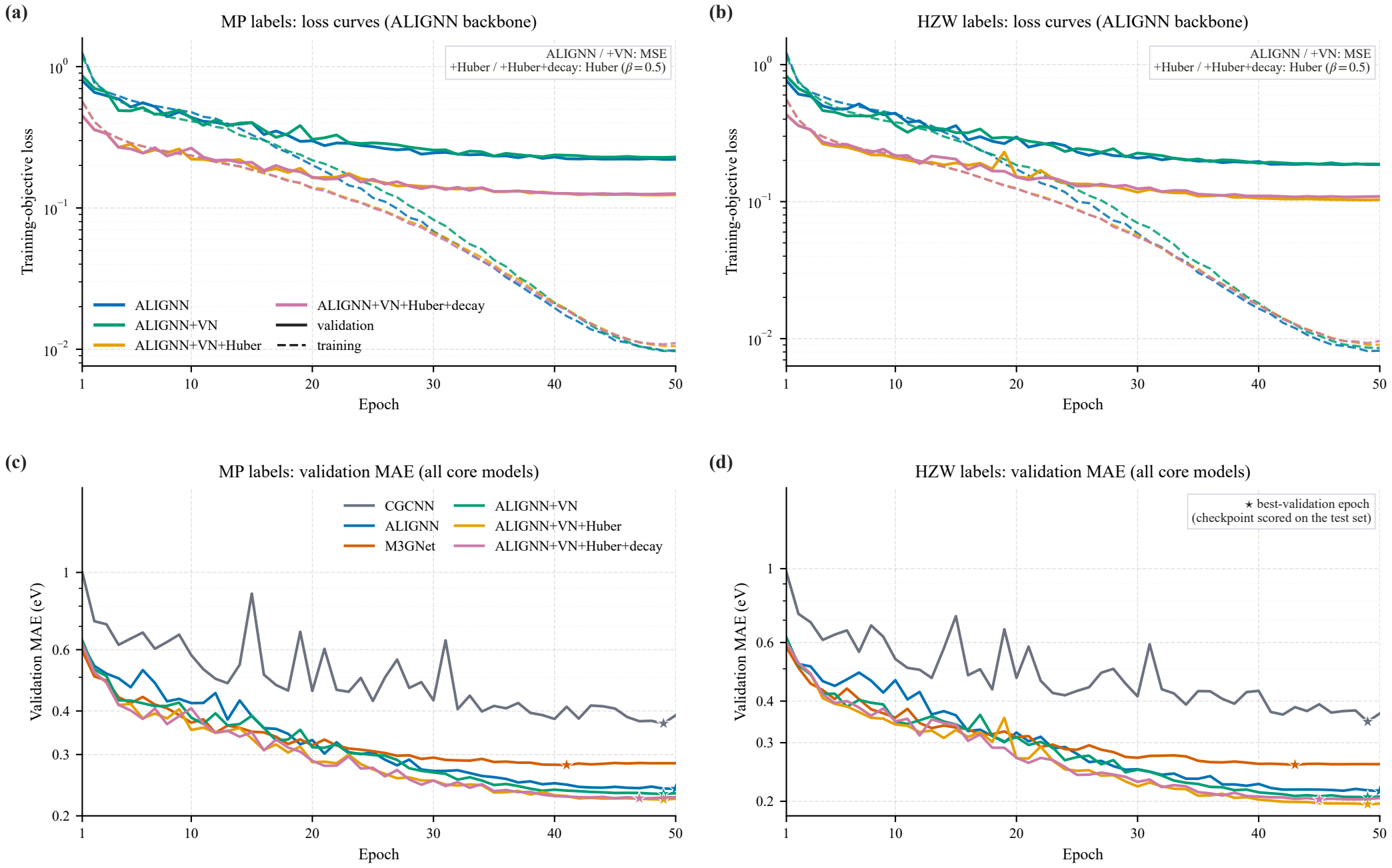

(a)
MP labels: loss curves (ALIGNN backbone)
ALIGNN / +VN: MSE
+Huber / +Huber+decay: Huber (β = 0.5)
Training-objective loss
Epoch
ALIGNN
ALIGNN+VN
ALIGNN+VN+Huber
ALIGNN+VN+Huber+decay
validation
training
(b)
HZW labels: loss curves (ALIGNN backbone)
ALIGNN / +VN: MSE
+Huber / +Huber+decay: Huber (β = 0.5)
Training-objective loss
Epoch
(c)
MP labels: validation MAE (all core models)
CGCNN
ALIGNN
M3GNet
ALIGNN+VN
ALIGNN+VN+Huber
ALIGNN+VN+Huber+decay
Validation MAE (eV)
Epoch
(d)
HZW labels: validation MAE (all core models)
★ best-validation epoch
(checkpoint scored on the test set)
Validation MAE (eV)
Epoch